

AP Three-Way Match — AI Workflow Audit

Client profile: mid-market manufacturing / distribution company **Workflow audited:** accounts-payable invoice processing (purchase-order three-way match) **Engagement type:** pre-build diagnostic (impact × risk scoring → automate / hybrid / manual) **Status:** proof-of-concept on synthetic data

This is a portfolio proof-of-concept. The dataset is synthetic and the dollar figures run on illustrative inputs. In a live engagement the rates and costs in `value_model.xlsx` are replaced with the client's own numbers, supplied by their FP&A and risk teams. The methodology, harness, and conclusions are the product.

1. Executive summary

The objective was not to ask "should we use AI in AP." It was to score each step of the AP workflow on business impact and on the cost of an AI error, then decide — per step — whether to automate, run hybrid (AI proposes, human approves), or leave manual.

The headline finding mirrors the well-documented industry pattern that most AI pilots fail on integration and risk, not model quality: **full automation of this workflow is the wrong answer, and it is wrong by millions.** The replay sandbox matched invoices at 99.4% overall accuracy, but caught only **50% of fraud (6 of 12 fraudulent invoices would have been paid).** Because a paid fraudulent invoice is irreversible cash out the door, automating the payment-release step destroys value even though the classifier looks excellent on average.

On illustrative inputs, the value model returns:

Operating model	Net annual value
Recommended (hybrid gates on fraud & payment release)	~\$3.28M
Naive full automation	~\$0.10M

The ~\$3.19M gap is the value of *not* over-automating. That is the engagement's single most important number.

2. Method

- Decompose** the single "process an invoice" workflow into eight atomic sub-steps that score differently on risk.
- Score** each on impact (1-5) and risk (1-5); the impact × risk position dictates the decision via a fixed matrix rule.
- Quantify** with a value equation: labor saved + losses prevented + discount capture – run cost – error cost.
- Measure risk empirically** by replaying a labeled dataset (including an over-sampled edge-case tail) through the candidate automation and recording where it fails — especially on the expensive, irreversible steps.

3. Sub-workflow scoring

Sub-workflow	Impact	Risk	Decision	Rationale
Invoice data capture (OCR/extract)	4	2	Automate	Deterministic; reversible before posting
Three-way match (clean invoices)	5	2	Automate	~81% of volume; high sandbox accuracy
Duplicate detection / auto-hold	4	2	Automate	100% recall; catches dupes humans miss
Price / quantity mismatch resolution	4	3	Automate*	Automatable with tolerance rules; sample-audit
Missing-PO / non-PO routing	3	3	Hybrid	Policy-dependent; route to an owner
Tax / GL coding	3	3	Hybrid	Automate with periodic sampling
Fraud / bank-change screening	5	5	Hybrid	Sandbox missed 50% of fraud; irreversible
Payment approval & release	5	5	Hybrid	Binding cash out; gate above a threshold

*Price/quantity resolution can begin hybrid and graduate to automate once the sandbox shows stable accuracy in shadow mode.

The two cells that carry the risk — fraud screening and payment release — are exactly the ones the matrix pulls into hybrid. AI drafts and flags; a human approves anything binding or irreversible.

4. Sandbox findings (empirical)

Replaying 1,000 labeled invoices (`harness/run_audit.py`):

- Overall accuracy:** 99.4%
- Auto-payable (clean) share:** 81.1% of volume
- Deterministic checks** (price, quantity, duplicate, missing-PO, tax): 100% recall

- **Fraud recall:** 50% — the residual miss is the entire argument for the human gate on payment release

The lesson is not "the model is bad." It is "the step where errors are most expensive is also the step where the automation is least reliable, and the error is irreversible." High average accuracy hides catastrophic tail risk.

5. Value model

See `value_model.xlsx` (three sheets: Assumptions, Workflow Audit, Value Model). All figures are formula-driven and recalculate when the client's real inputs are entered. On the illustrative inputs:

- Labor saved (hybrid): ~\$549K
- Duplicate losses prevented: ~\$940K
- Early-pay discount capture gain: ~\$1.94M
- Fraud losses incurred: \$0 (hybrid) vs ~\$3.24M (full automation)
- Cost to run: ~\$150K
- **Net annual value: ~\$3.28M (hybrid) vs ~\$0.10M (full automation)**

6. Recommendations

- Automate capture, clean matching, and duplicate auto-hold immediately.
- Keep fraud/bank-change screening and payment release **hybrid** with a hard human approval gate above a client-set dollar threshold (set by FP&A, not us).
- Start price/quantity and tax coding in hybrid; graduate to automate when shadow mode demonstrates stable accuracy.
- Instrument every step so the risk score is continuously re-measured, not assumed once.

7. De-risking rollout

Offline eval (this report) → shadow mode (engine runs on live traffic, actions disabled) → canary (acts on a small low-risk slice, human-reviewed) → full rollout with the hybrid gates above. Each gate has a numeric pass bar, e.g. "≥99% match accuracy and zero unreviewed payments above threshold."

8. Assumptions & limitations

- Synthetic data with an **elevated fraud rate (1.2%)** for demonstration; real AP fraud rates are far lower, but the conclusion holds because the case for the human gate rests on irreversibility and tail risk, not on expected value alone.
- Discount-capture and duplicate-prevention inputs are illustrative.
- The production automation would reach AP systems (ERP, vendor master, banking) through controlled, permissioned **MCP tools**. Every write-capable tool — "release payment," "update vendor bank detail" — is a risk surface and maps directly to the high-risk cells above; scoping those tool permissions is part of the audit, not an afterthought.

9. Roadmap

This is POC #1 of three, in ascending order of risk and governance:

1. **Manufacturing AP** (this) — clean hard-ROI baseline; establishes the harness.
2. **Finance / banking** — payments fraud and account takeover under AML/KYC.
3. **Healthcare (HIPAA)** — the same framework under strict PHI governance.